

6. Sieci komputerowe. Część 2 - Homework

1. Instalacja Docker Engine i Docker Compose.

Źródło: <https://docs.docker.com/engine/install/ubuntu/#install-using-the-repository>

Najpierw upewniłem się po moich wcześniejszych próbach instalacji Dockera Desktop na virtualnej maszynie Ubuntu 26.04 poprzez Oracle Virtual Box, że startuje od zera i nie mam żadnych pakietów, które utrudnią mi działanie od nowa z instalacją docker engine i docker compose:

sudo apt remove -y docker.io docker-compose docker-compose-v2 docker-doc podman-docker containerd runc (mam świadomość, że zwróci mi informację, że nie usunięto niektórych paczek, ponieważ nie były one zainstalowane)

```
vboxuser@Ubuntu:~$ sudo apt remove -y docker.io docker-compose docker-compose-v2 docker-doc podman-docker containerd runc
Note, selecting 'docker-compose-v2' instead of 'docker-compose'
Package 'docker.io' is not installed, so not removed
Package 'docker-compose-v2' is not installed, so not removed
Package 'docker-doc' is not installed, so not removed
Package 'podman-docker' is not installed, so not removed
Package 'containerd' is not installed, so not removed
Package 'runc' is not installed, so not removed
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 11
```

Sudo apt install -y ca-certificates curl: (Instaluje certyfikaty do połączeń HTTPS, curl pobieram dane)

```
vboxuser@Ubuntu:~$ sudo apt install -y ca-certificates curl
ca-certificates is already the newest version (20260601~26.04.1).
curl is already the newest version (8.18.0-1ubuntu2.4).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 11
```

- A) Tworze katalog, w którym będą przechowywane klucze repozytoriów APT.
- B) Pobieram oficjalny klucz GPG Dockera, którym APT będzie weryfikował pakiety
- C) Nadaje wszystkim użytkownikom prawo do odczytu klucza

```
vboxuser@Ubuntu:~$ sudo install -m 0755 -d /etc/apt/keyrings
vboxuser@Ubuntu:~$ sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
-o /etc/apt/keyrings/docker.asc
vboxuser@Ubuntu:~$ sudo chmod a+r /etc/apt/keyrings/docker.asc
vboxuser@Ubuntu:~$ echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

- D) Dodaje repozytorium Dockera z parametrami, że chce repozytorium w wersji “stable” (stabilne)

```
vboxuser@Ubuntu:~$ echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

sudo apt update: (ponownie odświeżam listę pakietów i tym razem Ubuntu zna również pakiety z repozytorium Dockera)

```
vboxuser@Ubuntu:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu resolute-security InRelease
Get:2 https://download.docker.com/linux/ubuntu resolute InRelease [32.5 kB]
Get:3 https://download.docker.com/linux/ubuntu resolute/stable amd64 Packages [23.7 kB]
Hit:4 http://pl.archive.ubuntu.com/ubuntu resolute InRelease
Hit:5 http://pl.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:6 http://pl.archive.ubuntu.com/ubuntu resolute-backports InRelease
Fetched 56.2 kB in 1s (98.4 kB/s)
11 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
(Zlecam instalacja docker engine, polecenie dockera, uruchamiania kontenerów, budowania obrazów, docker compose)

```
vboxuser@Ubuntu:~$ sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
Installing:
  containerd.io  docker-buildx-plugin  docker-ce  docker-ce-cli  docker-compose-plugin

Installing dependencies:
  docker-ce-rootless-extras  git  git-man  liberror-perl  pigz

Suggested packages:
  cgroupfs-mount | cgroup-lite  docker-model-plugin  git-doc  git-email  git-gui  gitk  gitweb  git-cvs  git-svn

Summary:
  Upgrading: 0, Installing: 10, Removing: 0, Not Upgrading: 11
  Download size: 108 MB
  Space needed: 423 MB / 30.8 GB available

Get:1 https://download.docker.com/linux/ubuntu resolute/stable amd64 containerd.io amd64 2.3.3-1~ubuntu.26.04~resolute [22.7 MB]
Get:2 http://pl.archive.ubuntu.com/ubuntu resolute/universe amd64 pigz amd64 2.8-1build1 [66.8 kB]
Get:3 http://pl.archive.ubuntu.com/ubuntu resolute/main amd64 liberror-perl all 0.17030-1 [23.5 kB]
Get:4 http://pl.archive.ubuntu.com/ubuntu resolute/main amd64 git-man all 1:2.53.0-1ubuntu1 [1,202 kB]
Get:5 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-cli amd64 5:29.7.2-1~ubuntu.26.04~resolute [17.0 MB]
Get:6 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce amd64 5:29.7.2-1~ubuntu.26.04~resolute [24.0 MB]
Get:7 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-buildx-plugin amd64 0.36.1-1~ubuntu.26.04~resolute [17.2 MB]
Get:8 http://pl.archive.ubuntu.com/ubuntu resolute/main amd64 git amd64 1:2.53.0-1ubuntu1 [4,227 kB]
Get:9 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-ce-rootless-extras amd64 5:29.7.2-1~ubuntu.26.04~resolute [10.2 MB]
Get:10 https://download.docker.com/linux/ubuntu resolute/stable amd64 docker-compose-plugin amd64 5.5.0-1~ubuntu.26.04~resolute [11.1 MB]
Fetched 108 MB in 3s (39.9 MB/s)
Selecting previously unselected package containerd.io.
(Reading database... 191924 files and directories currently installed.)
Preparing to unpack .../0-containerd.io_2.3.3-1~ubuntu.26.04~resolute_amd64.deb...
Unpacking containerd.io (2.3.3-1~ubuntu.26.04~resolute)...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a29.7.2-1~ubuntu.26.04~resolute_amd64.deb...
Unpacking docker-ce-cli (5:29.7.2-1~ubuntu.26.04~resolute)...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a29.7.2-1~ubuntu.26.04~resolute_amd64.deb...
Unpacking docker-ce (5:29.7.2-1~ubuntu.26.04~resolute)...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.8-1build1_amd64.deb...
Unpacking pigz (2.8-1build1)...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.36.1-1~ubuntu.26.04~resolute_amd64.deb...
Unpacking docker-buildx-plugin (0.36.1-1~ubuntu.26.04~resolute)...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5%3a29.7.2-1~ubuntu.26.04~resolute_amd64.deb...
Unpacking docker-ce-rootless-extras (5:29.7.2-1~ubuntu.26.04~resolute)...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../6-docker-compose-plugin_5.5.0-1~ubuntu.26.04~resolute_amd64.deb...
Unpacking docker-compose-plugin (5.5.0-1~ubuntu.26.04~resolute)...
Selecting previously unselected package liberror-perl.
Preparing to unpack .../7-liberror-perl_0.17030-1_all.deb...
```

ciąg dalszy...

```
Preparing to unpack .../8-git-man_1%3a2.53.0-1ubuntu1_all.deb...
Unpacking git-man (1:2.53.0-1ubuntu1)...
Selecting previously unselected package git.
Preparing to unpack .../9-git_1%3a2.53.0-1ubuntu1_amd64.deb...
Unpacking git (1:2.53.0-1ubuntu1)...
Setting up liberror-perl (0.17030-1)...
Setting up docker-buildx-plugin (0.36.1-1~ubuntu.26.04~resolute)...
Setting up containerd.io (2.3.3-1~ubuntu.26.04~resolute)...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' → '/usr/lib/systemd/system/containerd.service'.
Setting up docker-compose-plugin (5.5.0-1~ubuntu.26.04~resolute)...
Setting up docker-ce-cli (5:29.7.2-1~ubuntu.26.04~resolute)...
Setting up pigz (2.8-1build1)...
Setting up git-man (1:2.53.0-1ubuntu1)...
Setting up docker-ce-rootless-extras (5:29.7.2-1~ubuntu.26.04~resolute)...
Setting up docker-ce (5:29.7.2-1~ubuntu.26.04~resolute)...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' → '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' → '/usr/lib/systemd/system/docker.socket'.
Setting up git (1:2.53.0-1ubuntu1)...
Processing triggers for man-db (2.13.1-1build1)...
```

sudo systemctl enable --now docker (Uruchamianie Dockera, enable - będzie uruchamiał się przy starcie Ubuntu --now uruchomienie od teraz)

```
vboxuser@Ubuntu:~$ sudo systemctl enable --now docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
```

Sprawdzam czy Docker Engine działa “**sudo systemctl status docker**” - jego aktualny stan jest monitorowany, aż do zakończenia tego procesu

```
Aug 20 12:35
vboxuser@Ubuntu: ~ — sudo systemctl status docker

vboxuser@Ubuntu:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-08-20 12:34:07 UTC; 26s ago
     Invocation: 783dc516c2cd44bc9114f9af76f50801
   TriggeredBy: ● docker.socket
       Docs: https://docs.docker.com
     Main PID: 6392 (dockerd)
       Tasks: 10
      Memory: 134.1M (peak: 134.6M)
         CPU: 786ms
    CGroup: /system.slice/docker.service
            └─6392 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Aug 20 12:34:06 Ubuntu dockerd[6392]: time="2026-08-20T12:34:06.696831081Z" level=info msg="Restoring containers: start."
Aug 20 12:34:06 Ubuntu dockerd[6392]: time="2026-08-20T12:34:06.706320123Z" level=info msg="Deleting nftables IPv4 rules" error="running nft: /dev/stdin:1:17-30: Error: Could not process rule: No such file or directory\ndelete ta
Aug 20 12:34:06 Ubuntu dockerd[6392]: time="2026-08-20T12:34:06.805803216Z" level=info msg="Deleting nftables IPv6 rules" error="running nft: /dev/stdin:1:18-31: Error: Could not process rule: No such file or directory\ndelete ta
Aug 20 12:34:07 Ubuntu dockerd[6392]: time="2026-08-20T12:34:07.413114654Z" level=info msg="Loading containers: done."
Aug 20 12:34:07 Ubuntu dockerd[6392]: time="2026-08-20T12:34:07.436441584Z" level=info msg="Docker daemon" commit=ea43e3d containerd-snapshotter=true storage-driver=overlayfs version=29.7.2
Aug 20 12:34:07 Ubuntu dockerd[6392]: time="2026-08-20T12:34:07.436582976Z" level=info msg="Initializing buildkit"
Aug 20 12:34:07 Ubuntu dockerd[6392]: time="2026-08-20T12:34:07.545815648Z" level=info msg="Completed buildkit initialization"
Aug 20 12:34:07 Ubuntu dockerd[6392]: time="2026-08-20T12:34:07.556195559Z" level=info msg="Daemon has completed initialization"
Aug 20 12:34:07 Ubuntu dockerd[6392]: time="2026-08-20T12:34:07.556357726Z" level=info msg="API listen on /run/docker.sock"
Aug 20 12:34:07 Ubuntu systemd[1]: Started docker.service - Docker Application Container Engine.
[lines 1-23/23 (END)]
```

Testuje Dockera: "**sudo docker run hello-world**"

```
vboxuser@Ubuntu:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:5dd0d3e6e255913fc30f90b9f2b1d359cc2cbdb48090cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest
```

Hello from Docker!

This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:

1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
(amd64)
3. The Docker daemon created a new container from that image which runs the executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it to your terminal.

To try something more ambitious, you can run an Ubuntu container with:

```
$ docker run -it ubuntu bash
```

Share images, automate workflows, and more with a free Docker ID:

<https://hub.docker.com/>

For more examples and ideas, visit:

<https://docs.docker.com/get-started/>

Sprawdzam wersje Dockera Compose: "**docker compose version**"

```
vboxuser@Ubuntu:~$ docker compose version
Docker Compose version v5.5.0
```

Zadania dodatkowe

OpenSSL to zestaw narzędzi kryptograficznych służących między innymi do generowania kluczy, tworzenia i analizowania certyfikatów oraz weryfikowania podpisów. Wykonując zadania, korzystajcie z dokumentacji OpenSSL i samodzielnie znajdźcie polecenia oraz parametry potrzebne do osiągnięcia wskazanego rezultatu.

1. **Wygeneruj samopodpisany certyfikat TLS** dla domeny localhost i adresu 127.0.0.1. Certyfikat ma być ważny przez 365 dni i używać klucza RSA 2048 bitów oraz SHA-256.
2. **Wyświetl szczegółowe informacje o certyfikacie** i znajdź: właściciela, wystawcę, okres ważności, algorytm podpisu oraz Subject Alternative Name.
3. **Wyświetl informacje o kluczu prywatnym**, a następnie wyodrębnij z certyfikatu odpowiadający mu klucz publiczny. Wyjaśnij, który klucz można udostępniać.
4. **Zweryfikuj poprawność samopodpisanego certyfikatu**, traktując go jako zaufany certyfikat CA. Zapisz otrzymany rezultat i wyjaśnij, co oznacza.

Ad.1

Generuje dwa klucze z zadanymi parametrami:

```
openssl req -x509 -newkey rsa:2048 -sha256 -nodes -keyout localhost.key -out localhost.crt -days 365 -subj "/CN=localhost" -addext "subjectAltName=DNS:localhost,IP:127.0.0.1"
```

```
vboosuser@ubuntu:~$ openssl version  
OpenSSL 3.5.5 27 Jan 2026 (Library: OpenSSL 3.5.5 27 Jan 2026)  
  
vboosuser@ubuntu:~$ openssl req -x509 -newkey rsa:2048 -sha256 -nodes -keyout localhost.key -out localhost.crt -days 365 -subj "/CN=localhost" -addext "subjectAltName=DNS:localhost,IP:127.0.0.1"  
.....  
+-----+  
|               |  
|   +---+       |  
|   |         |  
|   |         |  
|   |         |  
|   +---+       |  
|               |  
+-----+
```

W ten sposób certyfikat jest ważny przez 365 dni, wykorzystuje RSA 2048 bitów i podpis SHA-256

Sprawdzam, czy pliki powstały:

```
ls -l localhost.key localhost.crt
```

```
vboxuser@Ubuntu:~$ ls -l localhost.key localhost.crt
-rw-rw-r-- 1 vboxuser vboxuser 1151 Aug 20 13:55 localhost.crt
-rw----- 1 vboxuser vboxuser 1704 Aug 20 13:55 localhost.key
```


Ad.2

openssl x509 -in localhost.crt -text -noout (pełne informacje o kluczu prywatnym)

```
vboxuser@Ubuntu:~$ openssl x509 -in localhost.crt -text -noout
Certificate:
  Data:
    Version: 3 (0x2)
    Serial Number:
      1b:82:f3:7d:01:96:cd:a6:f8:4f:69:9b:4b:0f:2a:8d:23:dc:12:36
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: CN=localhost
    Validity
      Not Before: Aug 20 13:55:11 2026 GMT
      Not After : Aug 20 13:55:11 2027 GMT
    Subject: CN=localhost
    Subject Public Key Info:
      Public Key Algorithm: rsaEncryption
      Public-Key: (2048 bit)
      Modulus:
        00:c3:7a:5d:67:b0:ce:d8:d3:00:b1:6c:90:a1:71:
        46:8c:91:c7:8b:48:13:cb:03:c8:12:55:79:a4:17:
        c3:57:11:ba:de:5d:22:bc:5c:40:bf:e8:d3:b2:2d:
        7e:20:8c:35:e3:04:97:83:71:56:ed:68:70:4d:a5:
        11:91:6b:12:59:51:38:88:a7:c5:6a:d5:f9:69:c6:
        3b:32:00:20:3f:e8:54:52:e1:f1:f3:11:91:17:f0:
        a8:68:64:e4:33:83:e8:0b:ec:3b:4a:5b:b0:49:50:
        6d:0b:73:3f:fd:d0:7c:4b:e3:ce:02:d9:ea:43:9c:
        8d:31:50:18:4b:a1:26:bf:76:90:dc:07:21:6c:b5:
        47:ed:18:73:95:ad:9f:c9:81:7b:bb:4d:66:d8:9e:
        ea:f6:05:11:76:37:56:21:26:df:3e:e2:8b:fb:36:
        fe:da:94:26:48:8b:00:67:34:05:20:4b:9c:95:bd:
        33:f2:26:89:76:fa:e7:61:b1:ce:73:ac:2d:48:34:
        ef:35:cb:f1:3d:62:17:03:49:72:e2:b9:62:0e:3c:
        51:90:42:bd:5c:d4:2b:ee:cf:d9:d7:2c:5b:22:b3:
        fa:0c:30:6c:77:17:66:7a:b5:61:78:57:21:a7:19:
        f9:26:8d:9d:c3:e3:8a:2c:fd:a4:72:5f:a0:15:4a:
        ac:61
      Exponent: 65537 (0x10001)
    X509v3 extensions:
      X509v3 Subject Key Identifier:
        C8:37:EA:AA:C8:B8:6C:C5:AD:05:94:2A:02:F2:2E:0B:23:75:82:FB
      X509v3 Authority Key Identifier:
        C8:37:EA:AA:C8:B8:6C:C5:AD:05:94:2A:02:F2:2E:0B:23:75:82:FB
      X509v3 Basic Constraints: critical
        CA:TRUE
      X509v3 Subject Alternative Name:
        DNS:localhost, IP Address:127.0.0.1
    Signature Algorithm: sha256WithRSAEncryption
    Signature Value:
      5d:be:8b:3f:ee:40:70:be:71:c3:70:a6:0b:25:58:42:82:ac:
      a6:2d:21:a3:c6:71:88:38:d5:05:7a:5e:61:a7:48:fb:08:95:
      50:ed:74:89:a0:b9:2d:ac:8f:8c:90:37:7a:27:66:d3:98:71:
      59:a2:59:73:77:11:1d:63:cb:82:04:30:4b:cd:ed:c5:a4:0c:
      82:2f:81:5e:a0:ab:cd:50:f4:a8:1d:c5:14:b9:d6:47:ba:e2:
      40:a2:12:26:5b:d7:cf:0b:d7:60:27:2a:60:b9:2b:a0:27:7b:
```

Właściciel (Subject):

Subject: CN = localhost

Wystawca (Issuer):

Issuer: CN = localhost

Okres ważności:

Validity

Not Before: ...

Not After : ...

(Not Before określa początek ważności, a Not After - koniec, 365 dni później)

Algorytm podpisu:

Signature Algorithm: sha256WithRSAEncryption

Subject Alternative Name:

X509v3 Subject Alternative Name:

"DNS:localhost, IP Address:127.0.0.1"

Ad.3

openssl rsa -in localhost.key -text -noout

```
vboxuser@Ubuntu:~$ openssl rsa -in localhost.key -text -noout
Private-Key: (2048 bit, 2 primes)
modulus:
 00:c3:7a:5d:67:b0:ce:d8:d3:00:b1:6c:90:a1:71:
 46:8c:91:c7:8b:48:13:cb:03:c8:12:55:79:a4:17:
 c3:57:11:ba:de:5d:22:bc:5c:40:bf:e8:d3:b2:2d:
 7e:20:8c:35:e3:04:97:83:71:56:ed:68:70:4d:a5:
 11:91:6b:12:59:51:38:88:a7:c5:6a:d5:f9:69:c6:
 3b:32:00:20:3f:e8:54:52:e1:f1:f3:11:91:17:f0:
 a8:68:64:e4:33:83:e8:0b:ec:3b:4a:5b:b0:49:50:
 6d:0b:73:3f:fd:d0:7c:4b:e3:ce:02:d9:ea:43:9c:
 8d:31:50:18:4b:a1:26:bf:76:90:dc:07:21:6c:b5:
 47:ed:18:73:95:ad:9f:c9:81:7b:bb:4d:66:d8:9e:
 ea:f6:05:11:76:37:56:21:26:df:3e:e2:8b:fb:36:
 fe:da:94:26:48:8b:00:67:34:05:20:4b:9c:95:bd:
 33:f2:26:89:76:fa:e7:61:b1:ce:73:ac:2d:48:34:
 ef:35:cb:f1:3d:62:17:03:49:72:e2:b9:62:0e:3c:
 51:90:42:bd:5c:d4:2b:ee:cf:d9:d7:2c:5b:22:b3:
 fa:0c:30:6c:77:17:66:7a:b5:61:78:57:21:a7:19:
 f9:26:8d:9d:c3:e3:8a:2c:fd:a4:72:5f:a0:15:4a:
 ac:61
publicExponent: 65537 (0x10001)
privateExponent:
 16:d7:11:d9:df:41:53:f9:57:8f:e3:91:04:a4:11:
 2b:b4:8e:dd:b9:d6:c2:36:57:66:b9:01:05:08:d7:
 c4:b1:28:25:fa:38:e8:4c:59:2e:73:3c:65:09:c6:
 be:73:d3:8a:0a:8b:cf:8a:6b:2f:43:30:38:0a:4d:
 02:b2:41:8b:19:1f:e3:6b:c2:f7:e8:37:83:c1:eb:
 41:bc:dd:07:ca:94:2b:b9:05:c5:b2:6c:43:21:55:
 f4:18:b1:c7:49:26:d0:59:d2:55:ae:5f:eb:39:0f:
 fc:b2:03:2c:9b:73:22:84:e4:19:d6:2f:76:a5:f0:
 a3:6d:e9:e0:d0:5c:10:2b:cc:79:18:32:29:65:40:
 c9:df:3d:20:fa:5c:f0:3c:17:1f:d8:c6:65:ea:70:
 cc:7f:06:d3:44:be:02:29:f0:58:3c:77:11:08:a5:
 46:68:5e:a4:36:1c:a7:af:5f:c0:93:9d:75:76:55:
 c4:cd:d0:fa:d1:31:25:94:83:fe:8c:67:c4:4a:a7:
 76:65:ee:f7:df:b2:73:3a:c7:c9:b1:a5:7b:d3:05:
 aa:f4:90:63:b3:f3:4c:88:b5:60:41:cb:33:3f:53:
 3a:e6:64:b8:01:d1:ac:c4:b6:a9:f8:1e:bd:34:e4:
 2b:c0:57:5a:6e:0f:f8:ec:bc:a8:78:85:af:9c:c5:
 c9
prime1:
 00:f0:52:fd:85:2c:89:dd:d0:e7:79:a5:46:10:92:
 a5:12:ae:1a:84:b5:79:f0:4f:b8:aa:7a:c1:71:33:
 72:df:e9:6c:78:a0:f9:31:3c:d0:eb:d5:29:79:66:
 98:f9:1f:63:80:16:12:9a:ea:a2:26:38:45:39:7b:
 dd:f8:58:d6:ec:28:31:67:09:cb:d7:90:02:22:61:
 27:10:e4:90:cd:a2:08:39:34:98:d6:5f:51:09:30:
 65:b3:4f:5e:fa:c0:72:59:4d:b6:fd:ff:d9:4b:86:
```

Private-Key: (2048 bit, 2 primes)

otrzymuje parametry klucza RSA, takie jak modulus, publicExponent i privateExponent

Klucz publiczny można wyodrębnić bezpośrednio z certyfikatu:

openssl x509 -in localhost.crt -pubkey -noout > localhost-public.key

("in" - input, dzięki temu wyodrębniłem klucz publiczny "-pubkey", "-text" - w wariantcie tekstowym, do pliku "localhost-public.key")

```
vboxuser@Ubuntu:~$ openssl x509 -in localhost.crt -pubkey -noout > localhost-public.key
```

Wyświetlam podgląd pliku "localhost-public.key" na ekranie:

cat localhost-public.key

```
vboxuser@Ubuntu:~$ cat localhost-public.key
-----BEGIN PUBLIC KEY-----
MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAw3pdZ7D02NMAswyQoXFG
jJHHi0gTywPIELv5pBfDVxG63l0ivFxAw+jTsi1+IIw14wSXg3FW7whwTaURkWsS
WVE4iKfFatX5acY7MgAgP+hUUuHx8xGRF/CoaGtKM4PoC+w7SluwSVBtC3M//dB8
S+POatnqQ5yNMVAYS6Emv3aQ3AchbLVH7RhZla2fyYF7u01m2J7q9gURdjdWISbf
PuKL+zb+2pQmSIsAZzQFIEucLb0z8iaJdvrnYbH0c6wtSDTvNcvxPWIXA0ly4rli
DjxRKEK9XNQr7s/Z1yxbIrP6DDBsdxdmerVheFchpxn5Jo2dw+OKLP2kcl+gFUqs
YQIDAQAB
-----END PUBLIC KEY-----
```

Klucz prywatny (localhost.key), nie wolno go udostępniać. Musi pozostać tajny.

Klucz publiczny, można bezpiecznie udostępniać. Po to jest.

Ad.4

Weryfikuje samopodpisany certyfikat:

openssl verify -CAfile localhost.crt localhost.crt

```
vboxuser@Ubuntu:~$ openssl verify -CAfile localhost.crt localhost.crt
localhost.crt: OK
```

Interpretacja:

"OK" oznacza, że OpenSSL uznał certyfikat za prawidłowy.

Nie oznacza to jednak, że localhost.crt jest automatycznie zaufany przez przeglądarkę lub system operacyjny. Jest to certyfikat samopodpisany.

Logika składni OpenSSL:

openssl verify [opcje] [certyfikaty]

verify

|

|— -CAfile localhost.crt

| ↓

| skąd wziąć zaufane CA

|

└─ localhost.crt



który certyfikat sprawdzić